Gahui Kim

CURRICULUM VITAE

AXIS Lab · Interaction Science Institute · Sungkyunkwan University
Seoul, South Korea · gahee020907@gmail.com · gahee020907.github.io · Google Scholar

RESEARCH PROFILE

Seeking a funded Doctoral Researcher or Research Assistant position in HCI, human–AI interaction, or immersive systems, with availability in 2027.

HCI researcher building and evaluating interactive AI systems in AR/XR. Uses controlled experiments and interaction logs to study how people interpret, reshape, adopt, or reject AI contributions, distinguishing changes in system output from changes in human thinking, agency, and experience.

RESEARCH AND PROFESSIONAL EXPERIENCE

AXIS Lab, Interaction Science Institute, Sungkyunkwan University — 2024–present
Undergraduate Research Assistant (2024–Feb 2025); Master's Student Researcher (Mar 2025–Aug 2026); Researcher (Sept 2026–present). Contributing to projects on proactive AI and human cognitive mechanisms in always-on AI environments, and continuing prototype-based research on AR smart glasses, socially visible sensing, and immersive human-AI interaction.

Researcher and App Developer, Personalized AI Story Generation Project — Apr 2026 – present
Developing Phodong, a production-stage SwiftUI/Firebase system that turns children's real-world object discoveries into personalised stories. Includes age-adaptive generation, parent-gated controls, and an instrumented discovery-to-story pipeline for a planned family study; supported by Seoul RISE and led by Prof. Changjun Lee.

Assistant Instructor, Generative AI Content Creation Workshop, Hyundai Futurenet — Jul 2026
Supported hands-on workshop activities on generative-AI content creation, including participant guidance and practical exercises for producing AI-assisted visual and interactive media.

Research Intern, RGB Makers Co., Ltd. — Aug 2025
Analysed first-person interaction data and privacy risks in metahuman-based content, developed a MetaHuman character in Unreal Engine 5, and proposed outdoor interaction scenarios for user-centred immersive experiences.

BK21 Research Assistant, Social Interaction Big Data Education and Research Group — Mar – Jun 2025
Supported immersive-device user research at AXIS Lab through literature review, study and protocol preparation, participant recruitment materials, lab setup, research ethics documentation, and data management.

EDUCATION

Sungkyunkwan University · Master of Science in Engineering, Immersive Media Engineering — Mar 2025 – Aug 2026
Advisor: Prof. Changjun Lee · AXIS Lab · Cumulative GPA: 4.45/4.50 (99.5/100)
Thesis: *When Sensing Becomes Socially Visible: Wearer Meaning and Adoption Resistance in AR Smart Glasses*

Sungkyunkwan University · Bachelor of Culture & Technology; Bachelor of Economics — Feb 2021 – Feb 2025
Major in Culture and Technology Convergence; International Trade Linked Major, Economics Track. Cumulative GPA: 4.32/4.50 (98.2/100). Graduated with SKKU Summa Cum Laude.

PUBLICATIONS

PEER-REVIEWED CONFERENCE PUBLICATIONS

Kim, G., Choi, Y., Kim, Y., & Lee, C. (2026). No Time to Say Goodbye: Emotional Loss Responses to Sudden Termination in Immersive AI Interactions. *CHI EA '26: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. ACM. DOI: 10.1145/3772363.3798687

Kim, G., Choi, Y., & Lee, C. (2025). See, Sense, Spark: Profiling Granularity in Proactive AI for Creative Research Ideation. *SIGGRAPH Asia 2025 Posters*. ACM. DOI: 10.1145/3757374.3771458

Kim, G., & Han, Y. (2024). Camp_Around: An Integrated Tour Service for International Students in Korea. *HCI Korea 2024 Creative Award Proceedings*, 1224–1227.

JOURNAL ARTICLES

Kim, G., Kim, S. Y., Park, S., Park, S., Choi, Y., & Lee, C. (2025). What Makes Audiences Pay More for Immersive Cinema? A Conjoint Analysis of ScreenX Preferences and Willingness-to-Pay. *International Telecommunications Policy Review*, 32(4), 1–45. DOI: 10.37793/ITPR.32.4.1

Kim, G., Choi, Y., & Lee, C. (2025). The Impact of Pre-adulthood Immersive Media Experiences on Continued Use Intention in Adulthood: An Empirical Study Applying the Value-based Adoption Model and Positive Technology Readiness Index. *Information Society & Media*, 26(1), 143–183.

WORKING PAPERS AND MANUSCRIPTS UNDER REVIEW

Kim, G., & Lee, C. (2026). Socially Visible Sensing in AR Smart Glasses: Outward Recording, Societal Concern, and Wearer Stigma. *Working paper.*

Kim, G., Choi, Y., & Lee, C. (under review). How Much Should Proactive AI Know About Its User? Profile Granularity in AI-Glasses Support for Research Ideation. *CHI 2027*. Manuscript under review.

Kim, G., Choi, Y., Kim, Y., & Lee, C. (under review). Goodbyes Matter: Immersive Relational Offboarding for Reducing Loss When AI Companion Interactions End. *CHI 2027*. Manuscript under review.

SELECTED FUNDED RESEARCH PROJECTS

Youth-Focused AI Risk Evaluation Benchmark Development — Researcher · ETRI · 2026
Proactive AI and Human Cognitive Mechanisms in Always-On AI Environments — Researcher · NRF · 2026–present
Personalized AI Story Generation for Children — Researcher · Seoul RISE · 2026–present
Interactive Experiential News Game Research and Development — Researcher · SKKU RISE · 2025–2026
Screen Multimodal Interaction Research · Big Data Analytics Team — Researcher · Samsung Electronics · 2024
Virtual-World Multi-Persona Tendencies and Cognitive Enhancement — Researcher · NRF Korea · 2025
Urban Green Exposure, Mobility, and Health — Researcher · AI Convergence Research Support · 2025
Psychological Counselling Chatbots: Humanities–Engineering Convergence — Researcher · NRF Korea · 2026
News and Social-Media Issue Analysis for the National Health Insurance Service — Researcher · NHIS · 2024

SELECTED RESEARCH AND PROJECTS · PORTFOLIO AND DEMOS

AR Smart Glasses and Socially Visible Sensing — Master's thesis, 2026
– Used a scenario-based experiment with 369 prospective users to compare outward recording, eye tracking, and AI analysis; wearer stigma was the only resistance factor directly associated with lower intention to use.
– Third study in a programme of work on AR smart glasses, alongside the hardware-based proactive-AI and AI-companion experiments below.

See, Sense, Spark — Proactive AI for Creative Research Ideation — SIGGRAPH Asia 2025 Poster
– Built a proactive AI system on XREAL Air 2 Pro. The poster study (analytic N = 17) found higher interest-driven curiosity with fine-grained profiles ($p = .016$, $d = 1.30$). The CHI 2027 follow-up (N = 26; 24 complete logs) found clear differences in AI-generated content, with no detectable differences in participants' question breadth or self-reports.

No Time to Say Goodbye — Termination and Emotional Loss in AI Companions — CHI EA 2026
– Built IRO and ran a between-subjects experiment on emotional loss (N = 26; ANCOVA, $p = .011$). The follow-up, Goodbyes Matter (CHI 2027, under review; N = 56), found that advance notice reduced immediate loss without reducing immersion.

ThinkCap — Critical Reading Interface — Independent Prototype in Development
– Prototyped a browser extension, native macOS overlay, and BLE firmware for line focus, critical marking, annotation, and behavioural logging; physical integration and evaluation remain in development.

Phodong: AI-Based Phygital Story Creation for Children — HCI Korea 2026 Creative Award Grand Prize
– Developing a SwiftUI/Firebase system that turns child-discovered objects into personalised stories and traces each object from discovery through story inclusion and child response.

Progressive AR Sound Cues — Research System in Development
– Implemented a DHH-informed Unity/XREAL package with separated event sources, cue policies, head-pose logs, and counterbalanced trials; device validation and participant studies remain pending.

SELECTED TECHNICAL DEVELOPMENT

3D Gaussian Splatting Reconstruction and Viewer — WebGL / GLSL, 2025
– Built a COLMAP/3D Gaussian Splatting workflow and reimplemented forward rasterisation, alpha blending, depth sorting, and camera-aware optimisation in JavaScript and GLSL.

Neural Models for Visual and Text Classification — PyTorch, 2026
– Evaluated an MLP for Fashion-MNIST and a padding-aware bidirectional LSTM for IMDb sentiment classification, comparing augmentation, regularisation, pooling, and training configurations.

Interactive Graphics and Geometry Processing — Three.js / WebGL / GLSL, 2025
– Implemented articulated animation and interactive graphics in Three.js, plus half-edge mesh processing, Catmull–Clark subdivision, and derivative-based texture filtering in GLSL.

AWARDS, EXHIBITIONS, AND CREATIVE WORK

HCI Korea 2026 Creative Award · Grand Prize for Phodong — 2026
Selected Team, 2026 Promising Student Startup Team 300+ · Growth Track — 2026
GS SHOP Content Leader Scholarship · 13th Cohort — Jan 2025
SKKU Summa Cum Laude · Sungkyunkwan University — Feb 2025
Best Award · 2nd Joint Academic Festival of Convergence Colleges — Aug 2024
Best Presentation Award · Cybercommunication Society Spring Conference — May 2024
Ars Electronica 2024 · Miracle on the Han River, co-creator and exhibitor, Linz — 2024
HCI Korea 2024 Creative Award Proceedings · Published contribution — 2024
Grand Prize · 2nd Culture and Technology Convergence Major Day Capstone Project Competition — Dec 2023

LEADERSHIP AND ACADEMIC SERVICE

President, 4th Student Council CNT:role, Culture and Technology Convergence — 2023
Led student community programmes at Sungkyunkwan University and helped launch the first joint academic festival across five convergence colleges.

RESEARCH SKILLS

Programming and data	Python; R; Swift; JavaScript; TypeScript; NumPy; pandas
AI, XR, and graphics	PyTorch; scikit-learn; Unity; Unreal Engine; XREAL; SwiftUI; Firebase; WebGL
Research methods	Controlled experiments; preregistration; interaction-log analysis; mixed models; permutation/bootstrap inference; SEM; qualitative coding
Research areas	Human–AI interaction; AI contribution and appropriation; proactive AI; agency and metacognition; AR/XR; trace-based evaluation
Languages	Korean (native); Chinese (proficient — six years of residence in China); English (academic reading, writing, and research communication); Finnish (beginner; currently studying)

REFERENCES

Prof. Changjun Lee · M.S. advisor · Sungkyunkwan University · changjunlee@skku.edu